

The EU Regulatory Patchwork for Dark Patterns

An Illustration of an Inframarginal Revolution in European Law?

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CONTENTS

- I Introduction
- II Dark Patterns and Their Impact
- III Patchwork of EU Regulatory Responses against Dark Patterns
 - A Triggers for Protection against Dark Patterns
 - 1 EU Consumer Law
 - 2 EU Data Protection Law
 - 3 Digital Services Act
 - 4 AI Act
 - 5 Data Act
 - 6 EU Competition Law
 - 7 Digital Markets Act
 - B Method of Enforcement
- IV Three Needs Resulting from the Current State of Play
 - A Need for Clarity on the Relationship between Legal Norms
 - B Need for Triggers for Protection Aligning with the Harm
 - C Need for Integrated Enforcement
- V Conclusion

I INTRODUCTION¹

In January 2023, the European Commission and a range of national consumer authorities in Europe announced that 148 out of 399 online shopping websites they had investigated as part of a screening exercise employed some form of dark

¹ Please note that this chapter was finalized in May 2023 and is based on sources available at that time. References to the AI Act and the Data Act were updated in January 2025 to reflect the adopted text, but other sources could not be updated or added.

patterns.² Dark patterns are ‘tricks used in websites and apps that make you do things that you didn’t mean to, like buying or signing up for something’.³ They concern manipulative practices that exploit consumer vulnerabilities or lead consumers to making choices that are not in their best interests.⁴

The announcement is only one instance of the current regulatory wave against dark patterns in the European Union (EU). Beyond the EU consumer rules,⁵ the General Data Protection Regulation (GDPR)⁶ can be relied upon to protect individuals against dark patterns in some circumstances. And scholarly literature has started to discuss the role of competition enforcement in this area as well.⁷ Beyond this, provisions targeting dark patterns have been included in the Digital Services Act,⁸ the Digital Markets Act,⁹ the Artificial Intelligence (AI) Act¹⁰ and the Data Act.¹¹ Each of these frameworks has its own substantive scope and enforcement mechanisms.

² Press Release European Commission, ‘Consumer protection: Manipulative online practices found on 148 out of 399 online shops screened’, 30 January 2023, available at https://ec.europa.eu/commission/presscorner/detail/en/ip_23_418.

³ The term ‘dark patterns’ was first used by Harry Brignull in 2010, available at [www.deceptive.design/#:~:text=Deceptive%20design%20patterns%20\(also%20known,companies%20that%20use%20these%20patterns](http://www.deceptive.design/#:~:text=Deceptive%20design%20patterns%20(also%20known,companies%20that%20use%20these%20patterns).

⁴ For a discussion of the definition of ‘dark patterns’, see OECD, ‘Dark Commercial Patterns’, OECD Digital Economy Papers October 2022 No. 336, pp. 8–9, available at www.oecd-ilibrary.org/docserver/44f5e846-en.pdf?expires=1678448863&id=id&accname=guest&checksum=6283E62D95CBE1A7A9E7A2A046E1B797.

⁵ Council Directive 93/13/EEC of 5 April 1993 on unfair terms in consumer contracts (Unfair Contract Terms Directive) [1993] OJ L 95/29; Directive 2005/29/EC of the European Parliament and of the Council of 11 May 2005 concerning unfair business-to-consumer commercial practices in the internal market (Unfair Commercial Practices Directive) [2005] OJ L 149/22; Directive 2011/83/EU of the European Parliament and of the Council of 25 October 2011 on consumer rights (Consumer Rights Directive) [2011] OJ L 304/64.

⁶ Regulation (EU) 2016/679 of the European Parliament and of the Council of 27 April 2016 on the protection of natural persons with regard to the processing of personal data and on the free movement of such data, and repealing Directive 95/46/EC (General Data Protection Regulation) [2016] OJ L 119/1.

⁷ See, for instance, Gregory Day & Abbey Stemler, ‘Are dark patterns anticompetitive’, *Alabama Law Review* 2020, Vol. 72 No. 1, pp. 1–45.

⁸ Regulation (EU) 2022/2065 of the European Parliament and of the Council of 19 October 2022 on a Single Market for Digital Services (Digital Services Act) [2022] OJ L 277/1.

⁹ Regulation (EU) 2022/1925 of the European Parliament and of the Council of 14 September 2022 on contestable and fair markets in the digital sector (Digital Markets Act) [2022] OJ L 265/1.

¹⁰ Regulation (EU) 2024/1689 of the European Parliament and of the Council of 13 June 2024 laying down harmonised rules on artificial intelligence and amending Regulations (EC) No 300/2008, (EU) No 167/2013, (EU) No 168/2013, (EU) 2018/858, (EU) 2018/1139 and (EU) 2019/2144 and Directives 2014/90/EU, (EU) 2016/797 and (EU) 2020/1828 (Artificial Intelligence (AI) Act) [2024] OJ L 2024/1689.

¹¹ Regulation (EU) 2023/2854 of the European Parliament and of the Council of 13 December 2023 on harmonised rules on fair access to and use of data and amending Regulation (EU) 2017/2394 and Directive (EU) 2020/1828 (Data Act) [2023] OJ L 2023/2854.

The increasing attention for the issue of dark patterns may be seen as an illustration of an inframarginal revolution in European law and policy. The use of dark patterns is typically an attempt by businesses to increase their revenue at the expense of consumers who purchase products they may not need, spend more time or give up more personal data than they would otherwise do. Furthermore, there is evidence that less educated consumers are more prone to be affected by dark patterns than more educated consumers.¹² Dark patterns thus raise distributive issues that are worth considering from an inframarginal perspective. The elimination of dark patterns has the simultaneous effect of redistributing wealth from firms to consumers and addressing inefficiencies that result from the distortions in demand. Dark patterns therefore implicate the part of inframarginalism involving redistribution of wealth that is efficient in character.

Instead of focusing on the more normative issue of when dark patterns should be considered harmful, the chapter compares the different legal frameworks applicable to these practices and asks to what extent the increasingly fragmented EU regulatory landscape can offer effective overall protection against dark patterns. The objective is to draw lessons for improving the enforcement against dark patterns based on the current EU regulatory framework and for informing possible future inframarginal revolutions against other practices in European law.

Section II gives some background to the issue of dark patterns. Section III analyses the different forms of legal protection available to address dark patterns by focusing on two features, namely the main triggers for protection (for instance, type of behaviour, processing of personal data, existence of market power) and the method of enforcement. Section IV evaluates the current state of affairs and offers some suggestions to improve the coordination between the different legal frameworks and enforcers involved. Section V concludes.

II DARK PATTERNS AND THEIR IMPACT

The term ‘dark patterns’ brings together a wide range of practices relating to the use of choice architectures or user interfaces to trick users into making decisions desired by the designer. Dark patterns can take different shapes and may, for instance, consist of pop-up boxes or the use of colours or persuasive language to give prominence to certain options. They can emerge in all sorts of services, such as apps, e-commerce platforms, cookie consent notices, online games, and so on.¹³ Beyond the variety in which they appear, what dark patterns have in common is that they often involve the exploitation of consumers’ behavioural biases.¹⁴ This includes

¹² Jamie Luguri & Lior Jacob Strahilevitz, ‘Shining a light on dark patterns’, *Journal of Legal Analysis* 2021, Vol. 13 No. 1, p. 47.

¹³ OECD, ‘Dark Commercial Patterns’, OECD Digital Economy Papers October 2022 No. 336, p. 9.

¹⁴ Luguri & Strahilevitz, ‘Shining a light on dark patterns’, p. 44.

biases like default or status quo bias (the tendency to remain with the default choice), framing effects (bias towards choosing options that are presented as a gain rather than a loss) and social proof bias (the tendency to make choices that align with those of others). Dark patterns trigger ‘System 1 thinking’ to induce individuals to respond more quickly and unconsciously without activating the more slow and rational ‘System 2 thinking’.¹⁵ Several categories of dark patterns are distinguished, namely:¹⁶

- forced action: consumers being tricked into thinking registration is necessary or sharing more personal data about themselves and others than intended
- interface interference: hidden information, emotionally manipulative framing, preselection of options
- nagging: repeated requests to act favourably to the business, exploiting consumers’ limited willpower
- obstruction: making it difficult for users to reject or opt out of a service
- sneaking: hidden costs or unanticipated automatic renewal of subscriptions
- social proof: false or misleading testimonials or activity messages of other consumers
- urgency or scarcity: messages about low stock, high demand or the use of a countdown timer

The ultimate goal behind a business’s decision to use dark patterns is to increase one’s revenue by tricking consumers into buying products or services they may not need, buying more than they need, or, in case of advertising-based business models, spending more time and attention or giving up more personal data than they would otherwise do. Commercial pressure and market forces may sometimes lead businesses to use dark patterns if everyone else does and give rise to a ‘race to the bottom’ when consumers can or do not discipline the use of dark patterns by switching to businesses who do not employ such practices.¹⁷ The inability of the market to self-correct against dark patterns pleads for regulatory intervention.

Dark patterns can cause economic as well as non-economic harm for individuals. Economic harm consists of financial losses when consumers purchase products or services they would not have bought otherwise or spend more money than they

¹⁵ Daniel Kahneman, *Thinking, Fast and Slow* (Penguin Books, 2011).

¹⁶ See consolidated versions of existing dark pattern taxonomies in OECD, ‘Dark Commercial Patterns’, OECD Digital Economy Papers October 2022, No. 336, p. 53 (Annex B) and Luguri & Strahilevitz, ‘Shining a light on dark patterns’, p. 53 (table 1).

¹⁷ M. R. Leiser, ‘Dark patterns: The case for regulatory pluralism between the European Union’s consumer and data protection regimes’, in E. Kosta, R. Leenes & I. Kamara (eds.), *Research Handbook on EU Data Protection Law* (Edward Elgar, 2022), p. 242 and Stigler Committee on Digital Platforms, Final Report 2019, p. 87.

intended. Non-economic harm concerns the impact on the individual's privacy and independent decision-making as well as emotional distress in situations where dark patterns induce impulsive and addictive behaviour.¹⁸ When a dark pattern causes a consumer to spend unnecessary time, attention or energy, the non-economic harm can also take the form of an increased cognitive burden.¹⁹ Beyond harm at the level of individual consumers and citizens, dark patterns can also have a collective impact. While the impact of dark patterns on individual welfare has dominated the literature so far, their effect on collective welfare can also be pronounced in the form of weaker competition and less consumer trust.²⁰ Dark patterns can reduce price transparency, limit the ability of consumers to choose offerings that truly match their preferences, and make them lose faith in market forces and disengage.²¹ As a result of these effects, the mechanism of supply and demand may no longer deliver proper market outcomes.²²

Commercial practices, including in a brick-and-mortar context, often have a persuasive element, and distinguishing persuasion from manipulation can be difficult.²³ However, dark patterns are more effective and therefore potentially more harmful than equivalent practices in a brick-and-mortar setting because of their larger scale and more intrusive character.²⁴ While the increasing regulatory attentions shows a strong consensus that dark patterns are harmful, establishing the boundaries of legal protection remains a challenge. The determination of what is and what is not allowed under the current EU legal framework mainly stems from inherent restrictions in the scope of the different legal domains and not so much from a normative assessment of the risks of dark patterns – as we will see in the next section.

III PATCHWORK OF EU REGULATORY RESPONSES AGAINST DARK PATTERNS

The nature of dark patterns brings together several policy concerns, both at the level of individual and collective welfare – as indicated in Section II. These concerns

¹⁸ Arunesh Mathur, 'Dark patterns at scale: Findings from a crawl of 11K shopping websites', *Proceedings of the ACM on Human-Computer Interaction*, Vol. 3/CSCW, pp. 3–4.

¹⁹ Arunesh Mathur, Jonathan Mayer & Mihir Kshirsagar, 'What makes a dark pattern ... dark? Design attributes, normative considerations, and measurement methods', *Proceedings of the 2021 CHI Conference on Human Factors in Computing Systems*, p. 10.

²⁰ Mathur et al., 'What makes a dark pattern ... dark?', p. 11.

²¹ OECD, 'Dark Commercial Patterns', OECD Digital Economy Papers October 2022 No. 336, pp. 26–28. On consumer disengagement, see Luguri & Strahilevitz, 'Shining a light on dark patterns', p. 69.

²² See also George A. Akerlof & Robert J. Shiller, *Phishing for Phools* (Princeton University Press, 2016).

²³ In the context of advertising, see Ramsi Woodcock, 'The obsolescence of advertising in the information age', *The Yale Law Journal* 2018, Vol. 127 No. 8, pp. 2204–585.

²⁴ Mathur et al., 'What makes a dark pattern ... dark?', p. 9.

relate to values like privacy, autonomy and competition as well as to interests regarding the protection of our society and democracy against the steering of choices by commercial players. It is thus logical that the use of dark patterns can trigger the application of several legal regime in parallel. This section analyses the different forms of legal protection in the current EU regulatory framework against dark patterns along two features: the main triggers for protection and the method of enforcement.

A Triggers for Protection against Dark Patterns

1 EU Consumer Law

The EU consumer law framework can be used to target dark patterns, both through broad principle-based protections as well as specific provisions. As this section will show, the notion of ‘fairness’ is at the heart of protection under EU consumer law. Beyond broad provisions prohibiting ‘unfair practices’ and ‘unfair contract terms’, the more specific protections applicable to dark patterns in the EU consumer law framework can be seen as concrete interpretations of the more general concept of ‘fairness’ as well.

Most relevant is the Unfair Commercial Practices Directive,²⁵ which prohibits a commercial practice as unfair if it is contrary to the requirements of professional diligence and ‘materially distorts or is likely to materially distort the economic behaviour’ of the average consumer.²⁶ The latter is at issue when the use of a commercial practice appreciably impairs ‘the consumer’s ability to make an informed decision, thereby causing the consumer to take a transactional decision that he would not have taken otherwise’.²⁷ In its December 2021 Guidance on the Unfair Commercial Practices Directive, the European Commission interpreted the requirement of professional diligence as including a general principle for traders to ‘take appropriate measures to ensure that the design of their interface does not distort the transactional decisions of consumers’.²⁸ One could regard this as a duty for

²⁵ For a discussion of the application of the Unfair Commercial Practices Directive to dark patterns, see also M. R. Leiser & Mireille M. Caruana, ‘Dark patterns: Light to be found in Europe’s consumer protection regime’, *Journal of European Consumer and Market Law* 2021, Vol. 10 No. 6, pp. 245–48.

²⁶ Article 5(2) of Directive 2005/29/EC of the European Parliament and of the Council of 11 May 2005 concerning unfair business-to-consumer commercial practices in the internal market (Unfair Commercial Practices Directive) [2005] OJ L 149/22.

²⁷ Article 2(e) of the Unfair Commercial Practices Directive. This is known as the ‘transaction test’.

²⁸ Commission Notice – Guidance on the interpretation and application of Directive 2005/29/EC of the European Parliament and of the Council concerning unfair business-to-consumer commercial practices in the internal market OJ C 526/1, p. 101.

traders to do 'fairness by design' by avoiding that interfaces give rise to dark patterns and instead ensuring that the design of the online environment enables people to make choices in a fair way.²⁹ According to the Commission, international standards and codes of conduct for ethical design may inform what the standard of professional diligence entails in the area of interface design and no intention is required for a dark pattern to be captured by the prohibition of unfair commercial practices.³⁰ Because of the open-ended nature of the requirement of professional diligence, many manipulative practices distorting the consumers' decision-making could potentially qualify as an unfair commercial practice. Annex I to the Unfair Commercial Practices Directive blacklists practices that are to be regarded as unfair in all circumstances. This includes practices that can be labelled as dark patterns, for instance the use of false statements referring to the limited time period in which the product or service will be available.³¹

Apart from laying down a general definition of 'unfairness', the Unfair Commercial Practices Directive distinguishes between two main categories of unfair practices: misleading and aggressive ones.³² Again, a key element in the assessment is whether the misleading or aggressive commercial practice 'causes or is likely to cause the average consumer to take a transactional decision that he would not have taken otherwise'.³³ Beyond this, a commercial practice is regarded as misleading if 'it omits material information that the average consumer needs' or contains false information, including overall presentation, and therefore 'deceives or is likely to deceive the average consumer' even if the information is factually correct.³⁴ The inclusion of the overall presentation of information as a possible factor is especially relevant for dark patterns involving hidden information or pre-selection and more prominent display of certain options.³⁵ The use of ambiguous language or trick questions may also qualify as a misleading practice or omission when it can deceive consumers.³⁶

²⁹ See also Netherlands Authority for Consumers and Markets (ACM), 'Guidelines on the protection of the online consumer', version of 15 March 2023, available at www.acm.nl/sites/default/files/documents/2020-02/acm-guidelines-on-the-protection-of-the-online-consumer.pdf.

³⁰ Commission Notice – Guidance on the interpretation and application of Directive 2005/29/EC of the European Parliament and of the Council concerning unfair business-to-consumer commercial practices in the internal market OJ C 526/1, p. 101.

³¹ No. 7 of Annex I to the Unfair Commercial Practices Directive.

³² Article 5(4) of the Unfair Commercial Practices Directive.

³³ Article 6(1), 7(1) and 8 of the Unfair Commercial Practices Directive.

³⁴ Article 6(1) (on misleading actions) and 7(1) (on misleading omissions) of the Unfair Commercial Practices Directive.

³⁵ BEUC, "Dark patterns" and the EU consumer law acquis – Recommendations for better enforcement and reform', February 2022, p. 7, available at www.beuc.eu/sites/default/files/publications/beuc-x-2022-013_dark_patterns_paper.pdf.

³⁶ Commission Notice – Guidance on the interpretation and application of Directive 2005/29/EC of the European Parliament and of the Council concerning unfair business-to-consumer commercial practices in the internal market OJ C 526/1, p. 101.

A commercial practice is aggressive if ‘by harassment, coercion, including the use of physical force, or undue influence, it significantly impairs or is likely to significantly impair the average consumer’s freedom of choice or conduct with regard to the product’.³⁷ Examples in the context of dark patterns are the use of emotions, for instance to make consumers feel guilty about not opting in to a service, or obstruction techniques whereby a consumer has to click through multiple steps in order to cancel a service.³⁸ Regarding the latter practices, the Commission required traders in its 2021 Guidance to adhere to the principle that unsubscribing from a service should be as easy as subscribing to the service.³⁹

Beyond the Unfair Commercial Practices Directive, the Consumer Rights Directive⁴⁰ and the Unfair Contract Terms Directive⁴¹ can offer protection against dark patterns in some circumstances. The Consumer Rights Directive contains a specific provision entitling consumers to reimbursement of payment for additional services outside of the main contractual obligation when the consent of the consumer has been obtained by the use of default options.⁴² In addition, the Consumer Rights Directive requires online marketplaces to provide the consumer with ‘general information, made available in a specific section of the online interface that is directly and easily accessible from the page where the offers are presented, on the main parameters determining ranking’.⁴³ More generally, the Consumer Rights Directive lays down information obligations that are applicable before the conclusion of a contract,⁴⁴ whereas the Unfair Contract Terms Directive focuses on the contract terms themselves and requires them to ‘be drafted in plain, intelligible language’.⁴⁵ As such, dark patterns relating to the lack of transparency or clarity in contract terms or information may fall foul of these two legislative frameworks. However, as many types of dark patterns occur beyond contracts, the Unfair

³⁷ Article 8 of the Unfair Commercial Practices Directive.

³⁸ Regarding obstruction techniques, see, for instance, the work of the Norwegian Consumer Council that led Amazon to make it easier for consumers to cancel their Prime subscription. See Press Release Norwegian Consumer Council, ‘Amazon makes it easier to cancel Prime following complaints from European consumer organisations’, 1 July 2022, available at www.forbrukerradet.no/side/amazon-makes-it-easier-to-cancel-prime-following-complaints-from-european-consumer-organisations/.

³⁹ Commission Notice – Guidance on the interpretation and application of Directive 2005/29/EC of the European Parliament and of the Council concerning unfair business-to-consumer commercial practices in the internal market OJ C 526/1, p. 102.

⁴⁰ Directive 2011/83/EU of the European Parliament and of the Council of 25 October 2011 on consumer rights (Consumer Rights Directive) OJ L 304/64.

⁴¹ Council Directive 93/13/EEC of 5 April 1993 on unfair terms in consumer contracts (Unfair Contract Terms Directive) OJ L 95/29.

⁴² Article 22 of the Consumer Rights Directive.

⁴³ Article 6a of the Consumer Rights Directive, as amended by Directive (EU) 2019/2161 of the European Parliament and of the Council of 27 November 2019 [2019] OJ L 328/7.

⁴⁴ Articles 5 and 6 of the Consumer Rights Directive.

⁴⁵ Article 5 of the Unfair Contract Terms Directive.

Commercial Practices Directive is the most relevant instrument in the current EU consumer protection framework.

What can be concluded from this overview is that the regulatory approach against dark patterns in EU consumer law consists of a continuum from open-ended provisions protecting against unfairness of commercial practices and contract terms, to more narrow prohibitions of misleading and aggressive commercial practices, and to very targeted prohibitions banning specific practices deemed unfair in all circumstances (including the Annex to the Unfair Commercial Practices containing blacklisted practices and the prohibition to use default options for additional payments in the Consumer Rights Directive). This combination of techniques strikes a balance between the effectiveness of targeted bans and the flexibility of open-ended protections to capture new concerns or practices that the legislator has not considered yet. However, only a limited range of dark patterns falls within the scope of the specific bans, so that in practice a case-by-case analysis is often required to assess whether a particular manipulative practice is to be considered as unfair under the more general provisions of the Unfair Commercial Practices Directive.

2 EU Data Protection Law

As the main instrument of EU data protection law, the GDPR⁴⁶ can offer protection against dark patterns if they involve the processing of personal data.⁴⁷ In February 2023, the European Data Protection Board (EDPB) published Guidelines on 'deceptive design patterns in social media platform interfaces', in which specific examples of dark patterns are described throughout the life cycle of a social media account and reference is made to data protection principles as they apply to dark patterns more generally.⁴⁸ There are several ways in which dark patterns may violate the GDPR. Compared to the EU consumer law framework, the protections available for dark patterns in the GDPR are not as targeted as, for instance, the specific bans on certain practices in the Unfair Commercial Practices Directive and the Consumer Rights Directive. Dark patterns relating to pre-ticked boxes will not meet the conditions for a valid consent in the GDPR and the ePrivacy Directive,

⁴⁶ Regulation (EU) 2016/679 of the European Parliament and of the Council of 27 April 2016 on the protection of natural persons with regard to the processing of personal data and on the free movement of such data (General Data Protection Regulation) [2016] OJ L 119/1.

⁴⁷ Article 4(1) of the GDPR defines personal data as: 'any information relating to an identified or identifiable natural person ("data subject"); an identifiable natural person is one who can be identified, directly or indirectly, in particular by reference to an identifier such as a name, an identification number, location data, an online identifier or to one or more factors specific to the physical, physiological, genetic, mental, economic, cultural or social identity of that natural person'.

⁴⁸ European Data Protection Board, 'Guidelines 03/2022 on Deceptive design patterns in social media platform interfaces: How to recognise and avoid them', 14 February 2023.

requiring an affirmative action from the data subject.⁴⁹ Similarly, dark patterns may involve the use of personal data for purposes incompatible with the original purposes of data collection (GDPR principle of purpose limitation) or go beyond what is necessary for the purposes for which the data are processed (GDPR principle of data minimization).⁵⁰ As a result, dark patterns may violate the regular EU data protection rules that are not specifically targeted at manipulative practices but aim to regulate the processing of personal data more generally. A similarity between the EU consumer and data protection frameworks is the role of the notion of ‘fairness’ as an underlying standard of protection.⁵¹

The principle of fair processing is key to the GDPR⁵² and requires personal data not to ‘be processed in a way that is unjustifiably detrimental, unlawfully discriminatory, unexpected or misleading to the data subject’.⁵³ According to the EDPB, the fairness principle acts as an umbrella that protects against ‘deceptive design patterns’ irrespective of whether other data protection principles are complied with.⁵⁴ The EDPB defines deceptive design patterns as: ‘interfaces and user journeys implemented on social media platforms that aim to influence users into making unintended, respectively unwilling, and/or potentially harmful decisions, often toward an option that is against the users’ best interests and in favour of the social media platforms interest, with regard to their personal data’.⁵⁵ Due to this broad definition, many dark patterns distinguished in Section II will fall foul of the GDPR’s principle of fair processing in the view of the EDPB as long as they involve the processing of personal data.

Other more general data protection principles relevant in the context of dark patterns are accountability, transparency and data protection by design.⁵⁶

⁴⁹ Article 4(11) of the GDPR and Articles 5 and 13 of Directive 2002/58/EC of the European Parliament and of the Council of 12 July 2002 concerning the processing of personal data and the protection of privacy in the electronic communications sector (ePrivacy Directive) [2002] OJ L 201/37. For a discussion of consent banners, see Colin Gray, Cristiana Santos, Nataliia Bielova, Michael Toth & Damian Clifford, ‘Dark patterns and the legal requirements of consent banners: An interaction criticism perspective’, *Proceedings of the 2021 CHI Conference on Human Factors in Computing Systems*, May 2021, available at <https://doi.org/10.1145/3411764.3445779>.

⁵⁰ Respectively, Articles 5(1)(b) and (c) of the GDPR.

⁵¹ For an in-depth discussion of the role of fairness in EU data protection law, see Damian Clifford & Jef Ausloos, ‘Data protection and the role of fairness’, *Yearbook of European Law* 2018, Vol. 37, pp. 130–187.

⁵² Article 5(1)(a) of the GDPR.

⁵³ European Data Protection Board, Guidelines 4/2019 on Article 25 Data Protection by Design and by Default, 20 October 2020, par. 69.

⁵⁴ European Data Protection Board, ‘Guidelines 03/2022 on Deceptive design patterns in social media platform interfaces: How to recognise and avoid them’, 14 February 2023, par. 3.

⁵⁵ European Data Protection Board, ‘Guidelines 03/2022 on Deceptive design patterns in social media platform interfaces: How to recognise and avoid them’, 14 February 2023, par. 9.

⁵⁶ European Data Protection Board, ‘Guidelines 03/2022 on Deceptive design patterns in social media platform interfaces: How to recognise and avoid them’, 14 February 2023, par. 10–19.

Accountability entails a responsibility of the data controller to ensure and demonstrate compliance with the GDPR, which can include tests or experiments showing how the design of the interface facilitates fair and informed choices relating to the processing of personal data.⁵⁷ The principle of transparency requires information relating to personal data processing to be provided to the data subject 'in a concise, transparent, intelligible and easily accessible form, using clear and plain language'.⁵⁸ Similar to the transparency and information requirements in EU consumer law, the GDPR thus has provisions protecting data subjects against a lack of clarity. Data protection by design concerns the implementation of 'appropriate technical and organisational measures . . . designed to implement data-protection principles . . . in an effective manner and to integrate the necessary safeguards into the processing' to meet the requirements of the GDPR and protect the rights of data subjects.⁵⁹ According to the EDPB, compliance with the principle of data protection by design would result in avoiding dark patterns in the first place as any problematic elements are to be identified before launching the service.⁶⁰

With its guidelines, the EDPB thus gives a strong signal that dark patterns violate the GDPR once they involve the processing of personal data and deceive data subjects. However, unless a dark pattern breaches a particular GDPR provision (for instance, the validity of consent in case of default options), a case-by-case assessment is required to determine whether the practice indeed manipulates the data subject's decision-making. Effective protection against dark patterns under the GDPR therefore also depends on adequate implementation and enforcement of the relevant data protection principles.

3 Digital Services Act

The Digital Services Act, adopted in October 2022, includes a provision requiring providers of online platforms not to 'design, organise or operate their online interfaces in a way that deceives or manipulates the recipients of their service or in a way that otherwise materially distorts or impairs the ability of the recipients of their service to make free and informed decisions'.⁶¹ While the provision is to be welcomed as a specific and clear prohibition on the use of dark patterns, it does have several limitations.

⁵⁷ Article 5(2) of the GDPR.

⁵⁸ Articles 5(1)(a) and 12(1) of the GDPR.

⁵⁹ Article 25(1) of the GDPR.

⁶⁰ European Data Protection Board, 'Guidelines 03/2022 on Deceptive design patterns in social media platform interfaces: How to recognise and avoid them', 14 February 2023, par. 19. For a discussion of how to apply the requirement of data protection by design to dark patterns, see Leiser, 'Dark patterns: The case for regulatory pluralism between the European Union's consumer and data protection regimes', pp. 246–248.

⁶¹ Article 25(1) of the Digital Services Act. See also recital 67.

First, the ban only applies to providers of an online platform defined as ‘a hosting service that, at the request of a recipient of the service, stores and disseminates information to the public’.⁶² While the Digital Services Act applies some of its other obligations to the broader category of intermediary services, including, for instance, internet service providers and web-hosting services,⁶³ these latter services are not affected by the ban on dark patterns.

Second, the Digital Services Act entitles the Commission with the competence to issue guidelines on how the ban on dark patterns applies to specific practices and mentions three of them: (a) giving more prominence to certain choices when asking the recipient of the service for a decision; (b) repeatedly requesting that the recipient of the service make a choice where that choice has already been made, especially by presenting pop-ups that interfere with the user experience; and (c) making the procedure for terminating a service more difficult than subscribing to it.⁶⁴ While these are indeed common dark patterns, other dark patterns also exist that seem equally problematic. If it is the wish of the EU legislator to prioritize these forms of manipulative practices, one wonders why it has not laid down stricter rules or even targeted bans against these practices instead of leaving it up to the Commission to issue guidelines.

Third, the ban on dark patterns contained in the Digital Services Act does not apply to practices covered by the Unfair Commercial Practices Directive or the GDPR.⁶⁵ As discussed in the previous sections, these two instruments can indeed target dark patterns but often require a case-by-case analysis, and because of the broad, principle-based protections, their scope is not clear-cut. Although the assumption of the exclusion in the Digital Services Act seems to be that effective protection is already available under the Unfair Commercial Practices Directive and the GDPR, this is thus not necessarily the case.

Compared to the more general fairness-related triggers for protection under EU consumer and data protection law, the Digital Services Act is more explicit by including a specific ban on dark patterns targeted at online platforms. However, as discussed, the scope of the ban is narrow so that the Unfair Commercial Practices Directive and the GDPR are likely to remain more relevant as instruments in protecting against dark patterns. At the same time, a welcome aspect of the ban on dark patterns contained in the Digital Services Act is that it goes beyond the individual relationship between a trader or data controller and the consumer or data subject that forms the heart of the assessment in EU consumer and data protection law. Even if dark patterns do not have clearly distinguishable harm at the level of individual consumers or data subjects due to lack of proof of financial loss or

⁶² Article 3(i) of the Digital Services Act.

⁶³ Article 3(g) of the Digital Services Act.

⁶⁴ Article 25(3) of the Digital Services Act.

⁶⁵ Article 25(2) of the Digital Services Act.

psychological detriment, they can still negatively impact the collective welfare by weakening competition and lowering consumer trust and engagement in markets.⁶⁶

The stronger focus on the more collective impact is also clear from another provision in the Digital Services Act, which requires providers of very large online platforms and search engines to assess systemic risks 'stemming from the design or functioning of their service and its related systems' including 'the design of their recommender systems and any other relevant algorithmic system including algorithmic systems'.⁶⁷ Considering the references to the design of services, the risk assessments may relate to practices involving dark patterns as well.

4 AI Act

The AI Act was adopted in June 2024 with the aim of laying down harmonized rules across the EU for AI systems.⁶⁸ A number of AI practices are prohibited, including: 'the placing on the market, the putting into service or the use of an AI system' that:

- (a) 'deploys subliminal techniques beyond a person's consciousness' or
- (b) 'exploits any of the vulnerabilities of a natural person or a specific group of persons due to their age, disability or a specific social or economic situation'

with the objective or the effect of materially distorting the behaviour of a person or a group of persons in a manner that causes or is reasonably likely to cause that person, another person or group of persons significant harm.⁶⁹ It is clear that these prohibitions target dark patterns as part of AI systems. The AI Act defines an AI system as 'a machine-based system that . . . infers, from the input it receives, how to generate outputs such as predictions, content, recommendations, or decisions that can influence physical or virtual environments'.⁷⁰

Beyond the AI system as constituting the material scope of the prohibitions, the main trigger for protection against dark patterns is similar to the Digital Services Act and the Unfair Commercial Practices Directive that both also require some form of material distortion of an individual's decision-making. While the Digital Services Act also targets the collective impact of dark patterns, the AI Act focuses on harm at the level of the individual or a group of individuals only and therefore takes a step backwards to the approach of the Unfair Commercial Practices Directive and GDPR again. Beyond this, the final text of Article 5 of the AI Act refers to 'significant harm' – whereas the proposal for the AI Act specifically referred to 'physical or

⁶⁶ OECD, 'Dark Commercial Patterns', OECD Digital Economy Papers October 2022 No. 336, pp. 26–28.

⁶⁷ Article 34(1) and (2) of the Digital Services Act.

⁶⁸ Article 1 of AI Act.

⁶⁹ Article 5(1)(a) and (b) of the AI Act.

⁷⁰ Article 3(1) of the AI Act.

psychological harm'.⁷¹ The recitals do clarify that significant harms are understood as 'in particular having sufficiently important adverse impacts on physical, psychological health or financial interests'.⁷² The reference to physical and psychological health is welcome (despite being moved to the recitals), because it indicates that the protections in the AI Act go beyond the more economic harm or financial interests relating to an individual's decision-making that is at the heart of the relevant provisions in the Unfair Commercial Practices Directive and, to some extent, the Digital Services Act as well. However, because of the requirement of individual harm at the level of a person or a group of persons, the AI Act's prohibitions of dark patterns seem to contribute little to capturing practices that are not already within the scope of the Unfair Commercial Practices Directive.⁷³

5 Data Act

Another relevant EU legislative instrument is the Data Act, which was adopted in December 2023. The Data Act aims at facilitating access to and use of data, with a particular focus on Internet of Things devices.⁷⁴ Within that context, the Data Act contains a specific provision requiring third parties that receive data at request of a user to refrain from making the exercise of choices or rights regarding data access 'unduly difficult, including by offering choices to the user in a non-neutral manner, or by coercing, deceiving or manipulating the user, or by subverting or impairing the autonomy, decision-making or choices of the user, including by means of a user digital interface or a part thereof'.⁷⁵ Explicit references are made to dark patterns in the recitals, indicating in particular that it 'should be as easy for the user to refuse or discontinue access by the third party to the data as it is for the user to authorize access' and that users should not be nudged 'into decisions on data disclosure transactions'.⁷⁶ Compared to the different legal frameworks discussed so far, the provision regarding dark patterns contained in the Data Act is narrow in scope. It only applies in cases where data are transferred using the data access rights of the Data Act. The practices to which the Data Act refers can already be captured under other regimes that similarly focus on the impact of dark patterns on the individual, namely the Unfair Commercial Practices Directive and the GDPR. While the Data

⁷¹ Article 5(1)(a) and (b) of the proposal for a Regulation of the European Parliament and of the Council on laying down harmonised rules on Artificial Intelligence (Artificial Intelligence Act), 21 April 2021, COM(2021) 206 final.

⁷² Recital 29 of the AI Act.

⁷³ Michael Veale & Frederik Zuiderveen Borgesius, 'Demystifying the Draft EU Artificial Intelligence Act Analysing the good, the bad, and the unclear elements of the proposed approach', *Computer Law Review International: A Journal of Information Law and Technology* 2021, Vol. 22 No. 4, pp. 98–100.

⁷⁴ Article 1(1) of the Data Act.

⁷⁵ Article 6(2)(a) of the Data Act.

⁷⁶ Recital 38 of the Data Act.

Act does not expand the scope of existing bans of dark patterns, it forms a welcome reminder for data holders to design their interfaces in relation to data access in a fair manner as well.

6 EU Competition Law

The relevance of EU competition law in targeting dark patterns mainly lies in its ability to capture the competitive impact of such practices, which goes beyond the effects at the level of the individual consumer or data subject on which the EU consumer and data protection frameworks focus. Scholarship has already started to discuss the extent to which competition law can address dark patterns.⁷⁷ As already noted above in the context of the Digital Services Act, dark patterns may cause harm to collective welfare even if individuals do not perceive any harm to them personally.⁷⁸ Where the harm to collective welfare takes the form of harm to competition, there may be a role for competition law to step in. Consumer sovereignty and consumers' freedom of choice are relevant values deserving protection under competition law as well. By manipulating consumer decision-making, market players can shape consumer choices and control competition by steering demand in directions that further their own commercial interests. This may not only result in consumers being exploited, but also in competition being weakened.⁷⁹

From the perspective of EU competition law, dark patterns could be a form of abuse of dominance under Article 102 of the Treaty on the Functioning of the European Union (TFEU). A distinction is made between exclusionary and exploitative abuses. Exclusionary abuse concerns behaviour foreclosing competitors from the market and indirectly harming consumers because the reduced level of competition will lead to higher prices, lower quality or less choice of products and services. Exploitative abuse involves conduct that directly harms consumers through the imposition of excessive prices or unfair conditions. While competition authorities in Europe have so far prioritized exclusionary conduct, cases against exploitative abuses seem to be on the rise. The most prominent example is the case against Facebook (now Meta) by the German competition authority in which it held Facebook liable for abusing its dominance in the market for social networks by

⁷⁷ See Day & Stemler, 'Are dark patterns anticompetitive', pp. 1–45 and Lirio Barros, Timo Klein, Anastasia Shchepetova & Tim Hogg, 'The rise of dark patterns: Does competition law make it any brighter?', *Competition Law Journal* 2022, Vol. 21 No. 3, pp. 136–44.

⁷⁸ OECD, 'Dark Commercial Patterns', OECD Digital Economy Papers October 2022 No. 336, pp. 26–28.

⁷⁹ See Inge Graef, 'Consumer sovereignty and competition law: from personalization to diversity', *Common Market Law Review* 2021, Vol. 58 No. 2, pp. 479–80.

imposing unfair conditions on users.⁸⁰ Beyond this, concerns relating to excessive data extraction and personalized pricing are discussed in the literature as possible forms of exploitative abuse.⁸¹ These are categories of behaviour that could also be applied to determine if certain dark patterns violate Article 102 TFEU as an exploitative abuse. Where dark patterns as employed by a dominant firm have the effect of lowering competition and foreclosing competitors from the market, they may also qualify as an exclusionary abuse.⁸² The German *Facebook* case provides an illustration of this, as the German competition authority argued that the unfair conditions Facebook imposed did not only exploit users but also gave Facebook an unfair advantage over competitors who did not engage in such behaviour.⁸³ There is thus room to enforce against dark patterns under EU competition law as well. Beyond establishing abuse and anticompetitive effects, the main trigger for taking action is the existence of a dominant position. This is a difference with the other legal domains discussed so far, which all target market players irrespective of whether they are dominant.⁸⁴ In this regard, EU competition law can complement the other legal regimes by targeting behaviour that has a competitive impact on consumers and markets overall.

7 Digital Markets Act

The Digital Markets Act was adopted in September 2022 and lays down a set of obligations for so-called gatekeepers that offer core platform services.⁸⁵ The

⁸⁰ Case B6–22/16, *Facebook – exploitative business terms*, 6 February 2019. The Court of Justice endorsed the approach of the German competition authority to rely on the GDPR to establish an exploitative abuse under competition law; see Case C-252/21 *Meta Platforms v Bundeskartellamt*, ECLI:EU:C:2023;537.

⁸¹ See Viktoria H. S. E. Robertson, 'Excessive data collection: Privacy considerations and abuse of dominance in the era of big data', *Common Market Law Review* 2020, Vol. 57 No. 1, pp. 161–90 and Marco Botta & Klaus Wiedemann, 'To discriminate or not to discriminate? Personalised pricing in online markets as exploitative abuse of dominance', *European Journal of Law and Economics* 2020, Vol. 20 No. 3, pp. 381–404.

⁸² Barros et al., 'The rise of dark patterns: Does competition law make it any brighter?', p. 141.

⁸³ Case B6–22/16, *Facebook – exploitative business terms*, 6 February 2019.

⁸⁴ However, note that some have proposed to incorporate a market power analysis in the interpretation of the Unfair Commercial Practices Directive in EU consumer law as well. See Johann Laux, Sandra Wachter & Brent Mittelstadt, 'Neutralizing online behavioural advertising: Algorithmic targeting with market power as an unfair commercial practice', *Common Market Law Review* 2021, Vol. 58 No. 3, pp. 719–50.

⁸⁵ Article 3(1) of the Digital Markets Act defines a gatekeeper as an undertaking that: (a) has a significant impact on the internal market; (b) provides a core platform service which is an important gateway for business users to reach end users; and (c) enjoys an entrenched and durable position, in its operations, or it is foreseeable that it will enjoy such a position in the near future. Once the quantitative thresholds laid down in Article 3(2) of the Digital Markets Act are met, there is a presumption that the undertaking qualifies as a gatekeeper. In accordance with Article 2(2) of the Digital Markets Act, the following services are core platform services: online intermediation services, online search engines, online social

obligations aim to ensure the fairness and contestability of the digital sector⁸⁶ and include prohibitions of self-preferencing, restrictions on the combination of personal data as well as duties regarding portability and interoperability.⁸⁷ Once an undertaking qualifies as a gatekeeper, it will have to comply with the list of obligations regardless of whether any harm can be established. This is a difference from the abuse of dominance framework of Article 102 TFEU, where action against dominant firms (which is a lower threshold than gatekeeper) can only be taken after anticompetitive effects of conduct have been established.

There are no obligations in the Digital Markets Act directly targeting dark patterns. However, dark patterns are addressed in the context of the anti-circumvention measures of the obligations imposed on gatekeepers. Gatekeepers cannot engage in behaviour undermining effective compliance with the obligations of the Digital Markets Act, 'regardless of whether that behaviour is of a contractual, commercial or technical nature, or of any other nature, or consists in the use of behavioural techniques or interface design'.⁸⁸ This means that dark patterns can be addressed under the Digital Markets Act when they are linked to behaviour regulated by the other obligations. The scope of application of the Digital Markets Act towards dark patterns is thus limited. The focus is more towards collective welfare, considering the objectives of fairness and contestability that the Digital Markets Act pursues.

B *Method of Enforcement*

The previous subsection has illustrated that the different regulatory frameworks analysed can each be applied to target dark patterns. Nevertheless, compared to what is possible on the basis of the legal rules, enforcement actions have so far been limited. Together with the substantive scope of the regulatory frameworks, the method of enforcement matters to ensure effective protection.

The EU consumer and data protection law frameworks are enforced at the national level.⁸⁹ Both regimes rely on a one-stop-shop system, where the outcome reached in a case with cross-border relevance investigated at the national level

networking services, video-sharing platform services, number-independent interpersonal communications services, operating systems, web browsers, virtual assistants, cloud computing services and online advertising services.

⁸⁶ Article 1(1) of the Digital Markets Act.

⁸⁷ Articles 5–7 of the Digital Markets Act.

⁸⁸ Article 13(4) of the Digital Markets Act.

⁸⁹ Article 68(5) of the GDPR and Article 1 of Regulation (EU) 2017/2394 of the European Parliament and of the Council of 12 December 2017 on cooperation between national authorities responsible for the enforcement of consumer protection laws (CPC Regulation) [2017] OJ L 345/1.

applies across the entire EU. Under the GDPR, the national data protection authority of the main establishment of the company under investigation is automatically competent to act as the lead authority.⁹⁰ In its two-year review of the GDPR, the European Commission pointed at the problems the latter system has created due to the largest tech companies being established in countries like Ireland and Luxembourg that do not have strong data protection authorities with enough resources to investigate important cross-border cases.⁹¹ The enforcement system in consumer protection where the national consumer authorities collaborate in the Consumer Protection Cooperation Network and assign one authority as best placed to coordinate the case can therefore be seen as more effective.⁹² The Consumer Protection Cooperation Network already took action against dark patterns under EU consumer law.⁹³

EU competition law relies on parallel enforcement at the EU and national level.⁹⁴ Decisions of the European Commission are binding across the EU, while actions of national competition authorities only have an effect in their respective territories. There is thus no one-stop-shop system applicable to enforcement actions of national competition authorities, unlike in consumer and data protection law.

The Digital Markets Act centralizes enforcement at the European Commission,⁹⁵ while the Digital Services Act relies on a dual enforcement model. Very large online platforms and search engines are subject to enforcement by the European Commission, and enforcement against other providers takes place at the national level.⁹⁶ And finally, both the AI and Data Acts foresee enforcement to happen at the level of the EU Member States.⁹⁷

What one can conclude from this overview is that the legal frameworks targeting especially powerful undertakings are enforced at the EU level (namely the Digital Services Act and the Digital Markets Act, and to some extent the EU competition rules as well – although they can also be enforced at the national level), whereas the legal frameworks with a general scope of application rely on enforcement at the national level. Market players active EU-wide are often the more powerful players,

⁹⁰ Article 56(1) of the GDPR.

⁹¹ Communication from the Commission to the European Parliament and the Council: Data protection as a pillar of citizens' empowerment and the EU's approach to the digital transition – two years of application of the General Data Protection Regulation, 24 June 2020, COM(2020) 264 final, p. 6.

⁹² Article 17(2) of the CPC Regulation.

⁹³ Press Release European Commission, 'Consumer protection'.

⁹⁴ Articles 4 and 5 of Council Regulation (EC) No 1/2003 of 16 December 2002 on the implementation of the rules on competition laid down in Articles 81 and 82 of the Treaty [2002] OJ L 1/1.

⁹⁵ Articles 20–31 of the Digital Markets Act.

⁹⁶ Articles 49 and 65 of the Digital Services Act.

⁹⁷ Article 70 of the AI Act and Article 37 of the Data Act.

so that enforcement at the EU level is especially relevant. For the legal frameworks enforced at the national level, an effective one-stop-shop mechanism is key to ensure that national authorities do not need to duplicate efforts, and divergences across Member States are minimized.

Beyond the issue of the level at which enforcement takes place, the *ex ante* or *ex post* nature of the regulatory instrument is an important factor impacting enforcement. *Ex ante* interventions prescribe future behaviour, while *ex post* interventions look back at past behaviour. Enforcement of EU consumer, data protection and competition law takes place in an *ex post* manner. The relevant authorities or courts take up cases and investigate whether behaviour that is or has been taking place violates the rules. Even if enforcement happens *ex post*, the interventions can also have an impact *ex ante* on the future strategy of market players through the setting of precedents. In practice, the difference may sometimes be a matter of degree and also depend on the specificity of the legal obligation. The notions of fairness in the consumer and data protection frameworks as well as the abuse of dominance prohibition in competition law are broad and open-ended. The advantage of open-ended provisions is that this offers flexibility to apply the provisions to new circumstances not yet foreseen by the legislator. The disadvantage of open-ended provisions is that it takes time to investigate and reinterpret the rules for new circumstances.

The other legal frameworks considered in this chapter, namely the Digital Services Act, the AI Act, the Data Act, the Digital Markets Act, all have more specific provisions in place targeting dark patterns and banning certain behaviour in a more *ex ante* manner. Although these provisions still contain open notions in need of application or interpretation, like whether a practice deceives users or materially distorts their behaviour, they are much more prescriptive than the EU consumer, data protection and competition law frameworks and aim to steer behaviour of market players upfront. The advantage of specific, *ex ante* obligations is that they can allow for more timely intervention because of their targeted nature. The disadvantage of specific, *ex ante* obligations is that their scope needs to be set upfront and can therefore at some point become outdated when concerns move towards other issues or areas.

One can argue that the combination of more open-ended regulation with more specific legislative obligations strikes a good balance between, on the one hand, having regulation that is flexible and future-proof because it can be applied to new circumstances and, on the other hand, having *ex ante* obligations that are specific and allow for timely intervention against practices for which there is consensus that they are harmful. The question is whether the substantive rules and enforcement mechanisms indeed complement each other and can offer effective overall protection against dark patterns (see the comparison in Table 12.1). The next section will reflect on this.

TABLE 12.1 *Comparison of how different EU regulatory frameworks apply to dark patterns*

	Main scope and triggers for protection	Focus on individual or collective harm	Type of enforcement
EU consumer law	Fairness of commercial practices and contract terms applicable to consumers	Individual harm	Ex post National level
EU data protection law	Processing of personal data Fair processing	Individual harm	Ex post National level
Digital Services Act	Online platforms Design, organise or operate online interfaces: • in a way that deceives or manipulates the recipients of platform services, or • in a way that otherwise materially distorts or impairs the ability of the recipients to make free and informed decisions	More towards collective harm	Ex ante EU level for very large online platforms and search engines National level for other online platforms
AI Act	Placing on the market, putting into service or use of an AI system Deployment of subliminal techniques or exploitation of vulnerabilities Material distortion of a person's behaviour Causing significant harm	Individual harm	Ex ante National level
Data Act	Third parties receiving data from a user under the data access right Making the exercise of choices or rights regarding the data access right unduly difficult	Individual harm	Ex ante National level
EU competition law	Dominance Abuse and anticompetitive effects	Collective harm	Ex post EU and national level
Digital Markets Act	Gatekeeper Core platform service Anti-circumvention of the obligations	Collective harm	Ex ante EU level

IV THREE NEEDS RESULTING FROM THE CURRENT STATE OF PLAY

The overview of main triggers and methods of enforcement in the previous section illustrates the current piecemeal nature of the EU regulatory framework against dark patterns. While useful complementarities may arise when parallel sets of rules target different concerns or protect different values, there are also risks of inconsistencies that may lead to either under- or over-enforcement due to the fragmentation of the overall regulatory framework. This section submits that three needs result from the state of play and offers suggestions to improve the enforcement against dark patterns based on the current EU regulatory framework. The recommendations can also inform the use of regulatory techniques more generally for other practices or areas where inframarginal revolutions are or will be occurring.

A Need for Clarity on the Relationship between Legal Norms

It is not uncommon for several legal regimes to apply in parallel to a particular situation. Legal systems have approaches in place to address conflicts of norms. Several doctrines of interpretation exist to decide which legal norm prevails in case several legal norms apply to the same matter, including: the *lex specialis* doctrine, according to which a law governing a specific matter takes precedence over a law governing general matters; the *lex superior* doctrine, according to which the hierarchically superior law overrides the hierarchically inferior one; and the *lex posterior* doctrine, according to which more recent law prevails over older law.

In the case of dark patterns, conflicts of norms mainly exist in the fact that some legal frameworks impose stricter or more specific obligations than others. Emotionally manipulative framing may, for instance, trigger the application of the Unfair Commercial Practices Directive, the GDPR, as well as the AI Act. How can one determine which rules and which legal regime apply?

These legal frameworks are all part of EU secondary law, so that there is no difference in their hierarchy. It is also hard to claim that the AI Act as the most recent regime applicable to dark patterns automatically takes precedence over the other, older relevant legal frameworks, because each of them has a different focus and underlying objective. Even though they all apply to emotionally manipulative framing, they do not aim at regulating the same subject matter or concern. And for the same reason applying the *lex specialis* doctrine is also not intuitive, considering that one could possibly regard each of the three regimes as the most specific one depending on the perspective one takes. Because of its focus on AI systems, the AI Act could be seen as the one having the most specific material scope of application. However, the Unfair Commercial Practices Directive aims at protecting the consumer against unfair commercial practices, which is a more specific concern compared to the more general range of obligations under the AI Act. In turn, the GDPR could be seen as the most specific one

from the perspective of its focus on the processing of personal data. As a result, application of the traditional doctrines of conflicts of norms does not give clarity on which legal framework to apply.

Looking at the recitals to the AI Act, they explicitly mention that the rules laid down in the AI Act ‘should be without prejudice to existing Union law, in particular on data protection, consumer protection, fundamental rights ...’.⁹⁸

Such use of ‘without prejudice’ clauses does not indicate how the legislator intended different regimes to interact. More upfront clarity would therefore be welcome to understand the relationship between legal regimes that apply in parallel to practices like dark patterns. Without additional guidance, it is up to regulators and courts to decide on this, which takes time and could undermine the effectiveness of the overall regulatory framework. The Digital Services Act does clearly outline that its ban on dark patterns does not apply to practices covered by the Unfair Commercial Practices Directive or the GDPR.⁹⁹ However, because these regimes rely on broad principle-based protections, it is not so apparent which dark patterns fall within their reach. By making the scope of the ban in the Digital Services Act dependent on these two other regimes with unclear boundaries, the EU legislator therefore in the end still leaves it up to regulators and courts to decide which dark patterns the Digital Services Act covers.

B Need for Triggers for Protection Aligning with the Harm

A concern stemming from the fragmented regulatory framework is whether the different triggers for protection can still capture the core of the harm resulting from dark patterns. The analysis in Section III illustrates that the triggers for protection are not determined by the type or nature of the harm, but rather by the inherent scope of the relevant legal framework. For instance, the extent to which the GDPR can target dark patterns is dependent on whether a practice involves personal data and meets the principle of fair processing, and EU competition law is only relevant in targeting dark patterns when a dominant firm is involved whose practices give rise to anticompetitive effects. The same is happening in the newly adopted legal regimes, with the Digital Services Act addressing dark patterns on online platforms, the Digital Markets Act targeting dark patterns relating to compliance with its other obligations, the AI Act focusing on dark patterns in AI systems, and the Data Act addressing dark patterns in case users give third parties access to data in the context of Internet of Things devices. This approach assigns protection based on the legal definitions of services, whereas the harm is agnostic to these definitions and does not stop at their outer boundaries.

While the fact that several legal frameworks apply in parallel can also lead to over-enforcement, the concern regarding dark patterns is so far rather one of

⁹⁸ Recital 9 of the AI Act.

⁹⁹ Article 25(2) of the Digital Services Act.

under-enforcement. Different mechanisms of protection apply that each have a regulator competent to act, but the fragmented nature of the overall regulatory framework may leave some dark patterns unaddressed that slip through the cracks between legal regimes. If there is consensus that certain practices are so problematic that they should be banned, it seems vital to ensure that the triggers for protection align with the harm. In the case of dark patterns, the harm consists of different aspects covering both individual and collective welfare. One can even wonder if it would not be more effective in such situations to have a specific regime in place with dark patterns as the regulatory target, rather than spreading protection across a range of legal domains that each protect against part of the harm resulting from dark patterns.

To do so, one would need to overcome the scepticism expressed by Easterbrook in 1996 in his 'Cyberspace and the Law of the Horse' that still impacts the field of technology regulation today in determining how to regulate new technologies or practices that do not neatly fit within the categories of existing legal frameworks.¹⁰⁰ Easterbrook argues that there is no need for specialized legal rules to regulate cyberspace just as it would make no sense to create a separate body of law for regulating all activities relating to horses. In his view, 'the best way to learn the law applicable to specialized endeavors is to study general rules' and any effort to collect separate sets of rules into a 'Law of the Horse' 'is doomed to be shallow and to miss unifying principles'.¹⁰¹ However, the regulatory framework regarding dark patterns is so fragmented that even unifying principles are lacking, despite the fact that protection stems from regimes with a general scope of application. If Easterbrook's arguments regarding the 'Law of the Horse' no longer hold up, should one create some sort of 'Law of Dark Patterns'?¹⁰² The Digital Services Act could have been the framework to put in place rules specifically targeting dark patterns, but the EU legislator decided to defer to the Unfair Commercial Practices Directive and the GDPR instead of banning certain dark patterns more widely.

C Need for Integrated Enforcement

With substantive rules regarding dark patterns now being included in various EU legislative instruments, the question is how to make enforcement most effective considering the fragmented regulatory landscape. The creation of more integrated modes of enforcement will be key in this regard. Coordination networks already exist to enable cooperation between the EU and the national levels in each of the different legal regimes in the form of the EDPB, the Consumer Protection

¹⁰⁰ For a discussion, see Ronald Leenes, 'Of Horses and Other Animals of Cyberspace', *Technology and Regulation* 2019, pp. 2–3.

¹⁰¹ Frank H. Easterbrook, 'Cyberspace and the law of the horse', *University of Chicago Legal Forum* 1996, Vol. 1996, p. 207.

¹⁰² In the context of a 'Law of Data', see Inge Graef, 'Paving the way forward for data governance: A story of checks and balances', *TechReg* 2020, Vol. 2020, pp. 27–28.

Cooperation Network and the European Competition Network. Similarly, the new Acts also create such bodies, namely the European Board for Digital Services, the European Artificial Intelligence Board, the European Data Innovation Board and the high-level group for the Digital Markets Act. The latter one includes other coordination networks, including the EDPB, the European Competition Network and the Consumer Protection Cooperation Network,¹⁰³ which is a welcome development considering that the other bodies only bring together enforcers within the respective Act.

The challenge is how to create coordination across these bodies and the different legal domains they represent. At the minimum, joint priority setting is desirable to clarify responsibility and better target enforcement resources. For instance, if data protection authorities focus on e-commerce or dark patterns involving social proof, consumer authorities could prioritize social media or dark patterns involving obstruction. By jointly setting priorities, both under-enforcement can be prevented as well as duplication of resources and battles among enforcers about who will take up a case.

Beyond this, there could be joint policymaking to clarify the scope of the legal rules and their interaction to address the concern discussed in Section IV.A. An illustration showing that it is feasible to do this is the 2021 joint statement by the Competition & Markets Authority and the Information Commissioner's Office in the UK on competition and data protection law.¹⁰⁴ A more recent example is the joint work of the European Commission and the EDPB to create guidance on the interplay between the DMA and GDPR.¹⁰⁵

A next level of integration would be joint cases. While this may seem complex to achieve, the international networks in the areas of consumer protection, the International Consumer Protection and Enforcement Network (ICPEN), and data protection, the Global Privacy Enforcement Network (GPEN), have already successfully collaborated. Following intervention by consumer authorities within ICPEN, Google announced in May 2021 that app providers will have to indicate on the Google Play Store what personal data each app keeps and potentially shares about its users. GPEN endorsed this joint action.¹⁰⁶ This is a first inter-network collaboration in the domains of data protection and consumer law.

¹⁰³ Article 40(2) of the Digital Markets Act.

¹⁰⁴ Competition and data protection in digital markets: a joint statement between the CMA and the ICO, 19 May 2021, available at https://assets.publishing.service.gov.uk/government/uploads/system/uploads/attachment_data/file/987358/joint_CMA_ICO_Public_statement_-_final_V2_180521.pdf.

¹⁰⁵ Press Release European Commission, 'Commission services and EDPB will start joint work on guidance on the interplay between DMA and GDPR', 10 September 2024, available at https://digital-markets-act.ec.europa.eu/commission-services-and-edpb-will-start-joint-work-guidance-interplay-between-dma-and-gdpr-2024-09-10_en.

¹⁰⁶ Press Release ICPEN and GPEN, 'Google Play Store to require app providers to provide consumers with detailed information regarding data collection and use following growing international pressure', 27 May 2021, available at www.pcpd.org.hk/english/whatsnew/files/google_play2021.pdf.

The enforcers across the different legal regimes in the EU could take inspiration from these examples, even though they are still scarce, to integrate enforcement and maximize the benefits of the current fragmented EU regulatory framework by dividing responsibilities and teaming up.

V CONCLUSION

Dark patterns are in the midst of a regulatory wave in the EU. Beyond existing regimes in the areas of consumer, data protection and competition law, many of the new Acts include a provision regarding dark patterns. On the one hand, this serves as a welcome reminder to market players about the harm associated with dark patterns. On the other hand, it creates fragmentation by spreading protection as well as responsibility for enforcement across a range of legal regimes and regulators. While parallel application of different legal regimes can be helpful if it creates synergies across areas, in the case of dark patterns the risk is so far rather one of under-enforcement.

To make enforcement as effective as possible, there is a need to integrate enforcement and assign responsibility across enforcers. This could happen through joint priority setting, joint policymaking and joint cases. At least, joint priority setting is needed so as to spread enforcement resources effectively and enhance each other's scope of action. Joint policymaking is welcome to clarify the relationship between the different legal regimes and their scope, which is often interdependent – as the ban on dark patterns in the Digital Services Act illustrates. Joint cases are the most advanced level of integration, where enforcers could exchange expertise and issue a common decision where concerns overlap.

As a lesson for future lawmaking and inframarginal revolutions against other practices, the increasing regulatory attention for dark patterns in the EU shows that it may not necessarily be effective to add protections across many legal regimes. The risk of such an approach is that despite the many applicable legal rules, some practices remain unaddressed because the protection becomes too piecemeal. Every legal regime has its own legal definitions and material scope of application that determine its range of action, while the harms cross the boundaries between legal regimes and lead to broader, overarching concerns. Effective lawmaking would require aligning the triggers of protection with the harm, which may sometimes call for the adoption of a specific regime for the targeted practices that prevails over existing regulation.